

# From Atomic Jumps to Stress-Driven Hydrogen Transport

Symbolic calculations of transport coefficients from atomic-scale kinetic data

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## Missing layer in multiscale materials modelling workflow

### The computational limit of numerical random-walks

**The Status Quo:**



Determining **diffusion coefficient** usually relies on **computationally heavy numerical simulations** (like KMC or MD) that average random-walk trajectories over massive time scales

**The Inefficiency:** Every change in temperature, strain, or migration barrier requires running the entire numerical simulation from scratch.

**Endless Computing Loop**

**Bravais vs non-Bravais Lattices**

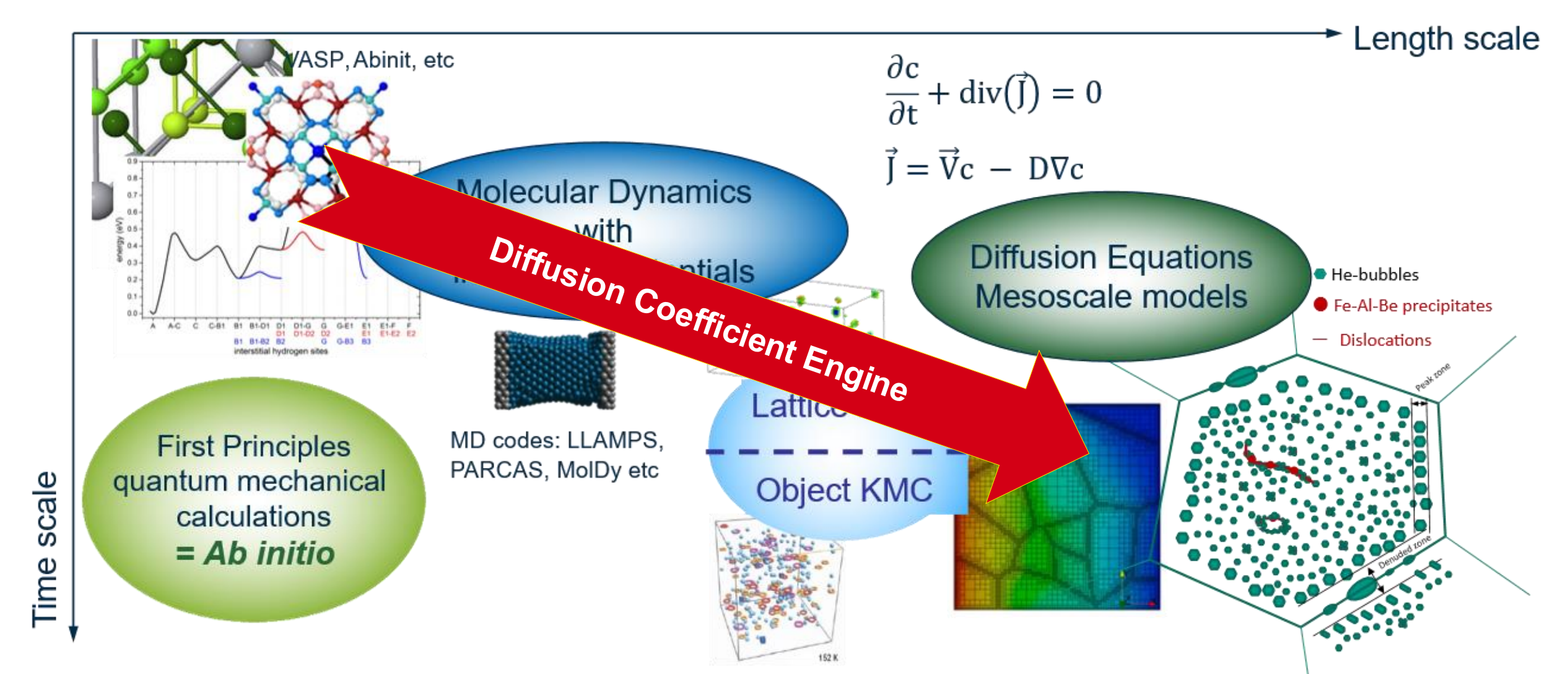
- Diffusion of vacancies or interstitials on Bravais lattice (with one site per unit cell) is **uncorrelated**. There is a simple analytical expression for calculating DC:

$$D_{ij}^{uncor} = \frac{1}{2} \sum_{\alpha=1}^n (h_i^{\alpha} \cdot h_j^{\alpha}) W(\vec{h}^{\alpha})$$

- Diffusion on non-Bravais lattices (with multiple sites per unit cell) is **mutually correlated** and requires solving a master equation:

$$\frac{dP_{\alpha}(t, t)}{dt} = \sum_{\beta=1}^N \sum_{l'} [W_{\beta\alpha}^{l'} P_{\beta}(l', t) - W_{\alpha\beta}^{l'} P_{\alpha}(l, t)]$$

### Bridging the gap in multiscale modelling workflow



## Automated evaluation of analytical transport coefficients

### Analytical Transport Parameters:

*Diffusion tensors and drift velocities from atomistic data*

- Diffusion tensor** is completely defined via its principal components, i.e., diffusion along principal axis.
- Ishioka and Koiwa reduced the 3D-diffusion to a 1D problem by projecting to a crystal direction perpendicular to a family of crystal planes.
- They obtained very compact representation of  $D_{\xi\xi}$ :

$$D_{\xi\xi} = -\frac{1}{2} (d_{\xi})^2 \left[ \frac{d^2}{d\theta^2} \det(\hat{A}(\theta)) \right]_{\theta=0} / \det(\hat{B})$$

where  $\hat{A}(\theta)$  is the Fourier transform of the transition matrix and  $\theta = (\vec{a}_1 \cdot \vec{\xi})t = d_{\xi} \cdot t$ , where  $d_{\xi}$  is the interplanar distance.

$\hat{B}$  is  $\hat{A}(\theta = 0)$  in which one row is replaced with ones.

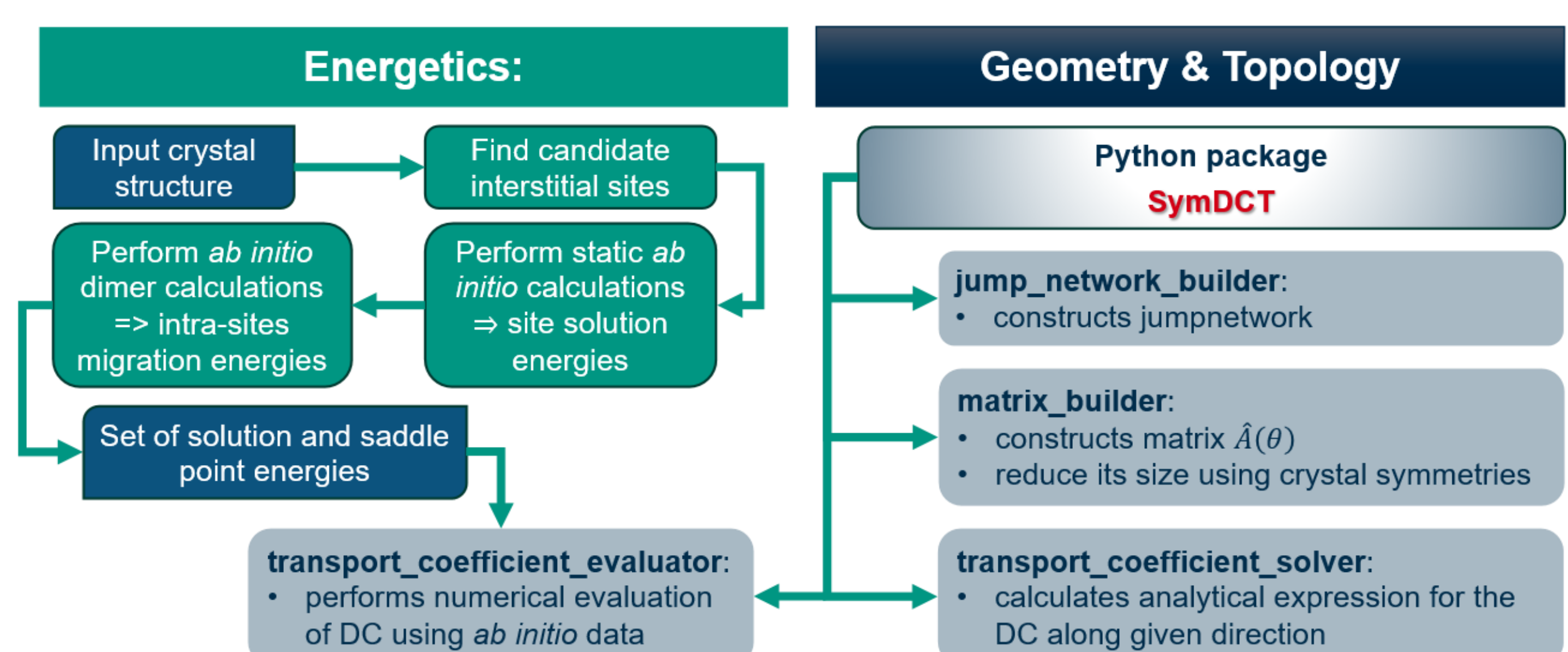
- Drift velocity** appears in transport equation if **external field** is applied.
- In this case, jumps **along** and **against** the external force are **not equivalent**
- Drift velocity** can be calculated similarly to the diffusion coefficient

$$v_{drift} = -i \frac{\partial}{\partial \theta} \det(\hat{A}(\theta)) \Big|_{\theta=-i\beta F} / \det(\hat{B})$$

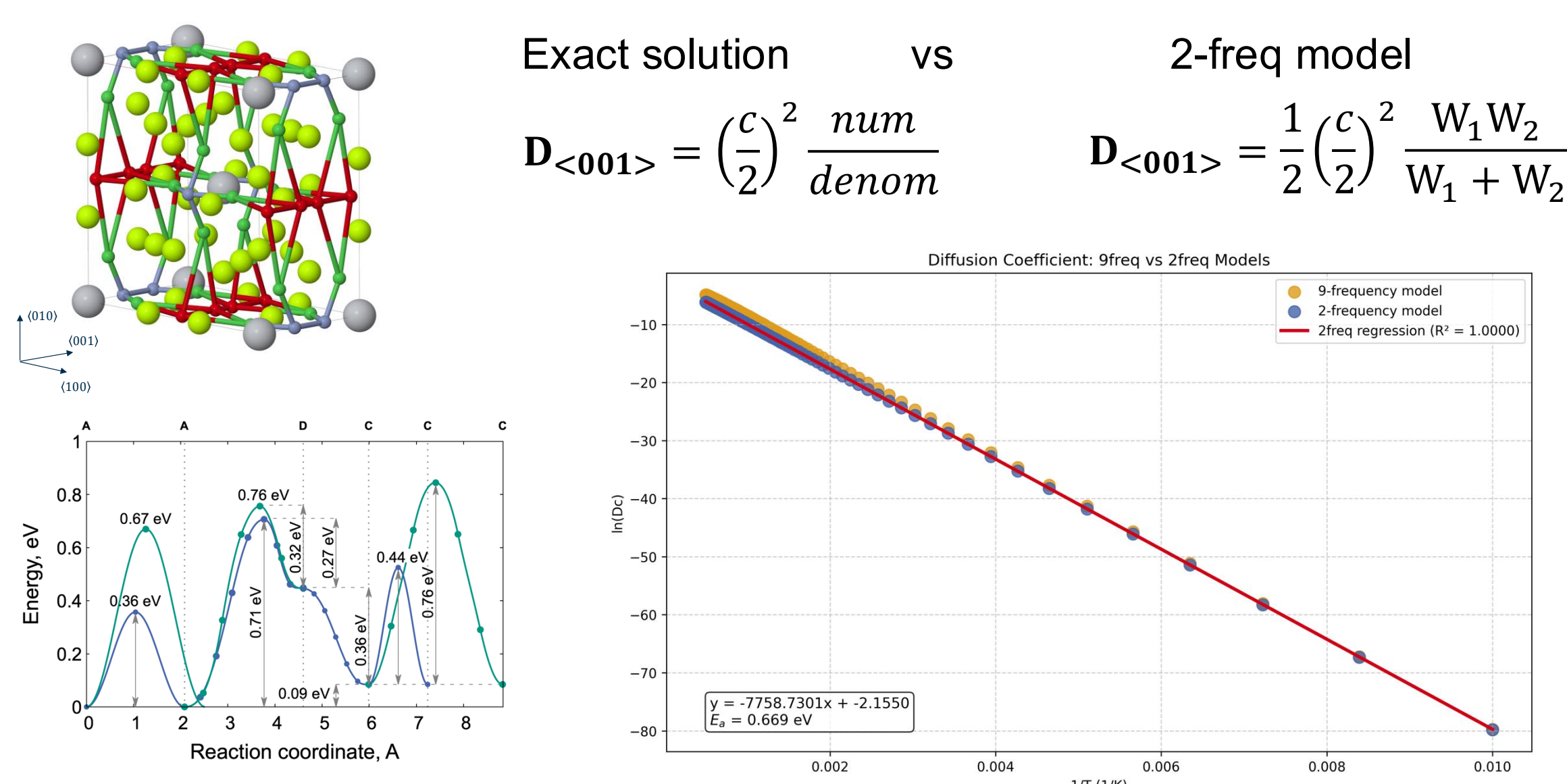
- If external field can be treated as perturbation  $\frac{F}{k_B T} \ll 1$ , **Einstein-Smoluchowski relation** for mobility  $\mu$  is restored

$$v_{drift} = \mu F \quad \mu = \frac{D}{k_B T}$$

### Resurrecting and automation of the Ishioka-Koiwa method



### Helium interstitial diffusion in Be<sub>12</sub>Ti



### Shifting from numerical averaging to analytical precision

|                              | Numerical Simulations                           | SymDCT Analytical Approach                                      |
|------------------------------|-------------------------------------------------|-----------------------------------------------------------------|
| <b>Methodology</b>           | Stochastic averaging over time and space        | Exact analytical expressions                                    |
| <b>Computational Expense</b> | High: Requires rerun for every parameter change | Near-zero: Evaluates the generated function instantly           |
| <b>Reusability</b>           | Specific to one rigid condition                 | Universal for any material with identical jump-network topology |
| <b>Complex Conditions</b>    | Struggles with inhomogeneous fields             | Natively supports liner-response analysis for external fields   |



Link to GitHub repository

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